

Random Signal Theory–EEE 554

Lecture Notes 3

Conditional Probability Density Functions

[Reading: Kay, Sections 8.3, 8.4, 8.6, 13.3, 13.4, 13.6]

Concept of joint conditional PDF for continuous RVs X and Y :

For small Δx , the probability of $Y \leq y$, given X is in a small neighbourhood of x , is

$$\begin{aligned} \Pr\left(Y \leq y \mid x - \frac{\Delta x}{2} \leq X \leq x + \frac{\Delta x}{2}\right) &= \Pr(Y \leq y \mid X = x) \\ &= \int_{-\infty}^y f_{Y|X}(u \mid x) \, du = F_{Y|X}(y \mid x) \end{aligned}$$

► Joint conditional PDF

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_{XY}(x, y)}{\int f_{XY}(x, y) \, dy}$$

$$\Rightarrow f_{XY}(x, y) = f_{Y|X}(y \mid x) f_X(x) = f_{X|Y}(x \mid y) f_Y(y)$$

► Bayes' relation for conditional PDFs

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y) f_Y(y)}{f_X(x)} = \frac{f_{X|Y}(x \mid y) f_Y(y)}{\int f_{X|Y}(x \mid y) f_Y(y) \, dy}$$

where the marginal is obtained as $f_X(x) = \int f_{X|Y}(x \mid y) f_Y(y) \, dy$

Example: Given joint PDF, $f_{X,Y}(x,y) = \begin{cases} 1.2(1 - x^2 y), & 0 < x \leq 1, 0 < y \leq 1 \\ 0, & \text{otherwise} \end{cases}$

Find the probability of $X \leq 0.5$, given that $Y = 0.25$

$$\Pr(X \leq 0.5 \mid Y = 0.25) = \int_0^{0.5} f_{X|Y}(u \mid 0.25) \, du = \int_0^{0.5} f_{X,Y}(u, 0.25) \, du$$

To find the conditional PDF, use $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$

Since joint PDF is given, find marginal PDF of Y , for $S_Y \in (0, 1)$

$$f_Y(y) = \int_0^1 1.2(1 - x^2 y) \, dx = \left(1.2x \Big|_0^1 \right) - \left(\frac{1.2}{3} x^3 y \Big|_0^1 \right) = 1.2 \left(1 - \frac{y}{3} \right), \quad 0 < y \leq 1$$

$$\Rightarrow f_{X|Y}(x \mid y) = \frac{(1 - x^2 y)}{(1 - y/3)}, \quad 0 < y \leq 1, 0 < x \leq 1$$

$$\Rightarrow \Pr(X \leq 0.5 \mid Y = 0.25) = \int_0^{0.5} f_{X|Y}(u \mid 0.25) \, du = \int_0^{0.5} \frac{(1 - u^2 (0.25))}{(1 - (0.25)/3)} \, du = 0.53$$

- If X and Y are independent

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_Y(y) f_X(x)}{f_X(x)} = f_Y(y)$$

and similarly $f_{X|Y}(x | y) = f_X(x)$

- Mean of conditional PDF

Since PDF depends on x , mean is also a function of x

$$E_{Y|X}[Y | x] = \int y f_{Y|X}(y | x) \, dy$$

Also, $g(X) = E_{Y|X}[Y | X]$ is a function of RV X

$$\begin{aligned} E_X[E_{Y|X}[Y | X]] &= E_X[g(X)] = \int (E_{Y|X}[Y | x]) f_X(x) \, dx \\ &= \int \left(\int y f_{Y|X}(y | x) \, dy \right) f_X(x) \, dx = \int y (f_{Y|X}(y | x) f_X(x)) \, dx \, dy \\ &= \int y \left(\int f_{XY}(x, y) \, dx \right) \, dy = \int y f_Y(y) \, dy = E_Y[Y] \end{aligned}$$

Optimal prediction makes use of mean of conditional PDF

prediction minimum mean-squared error (MSE) of Y given $X = x$ is $E_{Y|X}[Y | x]$

Consider two RVs, Y and X

observe $X = x$, want to predict a value of y , where $Y = g(X)$

To form MSE: squared error is $(Y - g(X))^2$ and mean by expectation $E[(Y - g(X))^2]$,

$$\begin{aligned}\text{MSE} &= \int \int (y - g(x))^2 f_{XY}(x, y) \, dx \, dy = \int \int (y - g(x))^2 f_{Y|Y}(y | x) f_X(x) \, dx \, dy \\ &= \int f_X(x) \left(\int (y - g(x))^2 f_{Y|Y}(y | x) \, dy \right) dx\end{aligned}$$

To minimize MSE, take derivative of inner integral and set derivative to zero

$$\frac{\partial}{\partial g(x)} \left(\int (y - g(x))^2 f_{Y|Y}(y | x) \, dy \right) = \int 2(y - g(x)) f_{Y|X}(y | x) \, dy = 0$$

$$\Rightarrow \text{MMSE: } \int y f_{Y|X}(y | x) \, dy = g(x) \int f_{Y|X}(y | x) \, dy = g(x)/r(x) \text{ and}$$

$$\hat{g}(x) = r(x) \int y f_{Y|X}(y | x) \, dy = r(x) E[Y | x]$$

Bayes' rule of one RV conditioned on an event

Consider CRV X , sample space $S_X = \mathbb{R}$; event A contains a subset of $x \in S_X$ values

CDF conditioned on event

$$F_X(x \mid A) = \Pr(X \leq x \mid A) = \frac{\Pr(X \leq x, A)}{\Pr(A)}$$

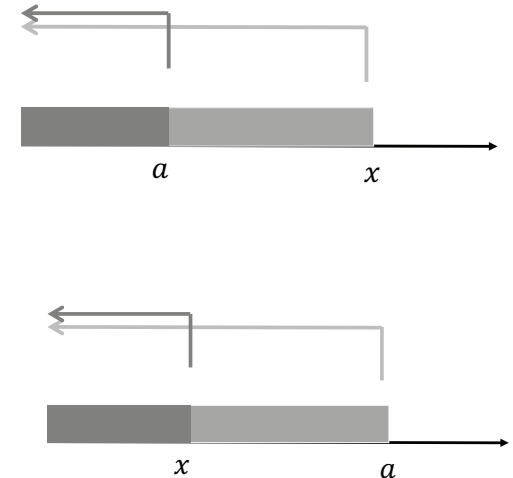
where $\Pr(X \leq x, A)$ is the joint probability or intersection of both events

Different cases of conditional CDF depending on the value of a

Case 1: $A = \{X \leq a\}, a \in S_x$

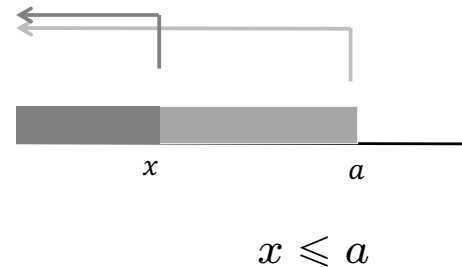
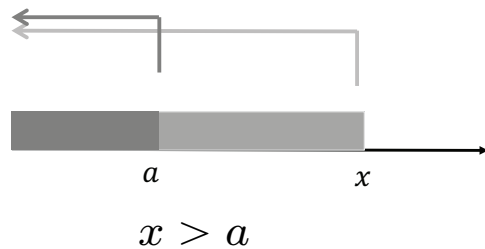
$$F_X(x | A) = F_X(x | X \leq a) = \Pr(X \leq x | X \leq a)$$

$$= \frac{\Pr(X \leq x, X \leq a)}{\Pr(X \leq a)} = \begin{cases} \frac{\Pr(X \leq a)}{\Pr(X \leq a)} = 1, & x \geq a \\ \frac{\Pr(X \leq x)}{\Pr(X \leq a)} = \frac{1}{F_X(a)} F_X(x), & x \leq a \end{cases}$$



$F_X(a) \geq F_X(x)$ when $x \leq a$ (divide by fraction less than one)

So, $A = \{X \leq a\}$



Condition CDF is

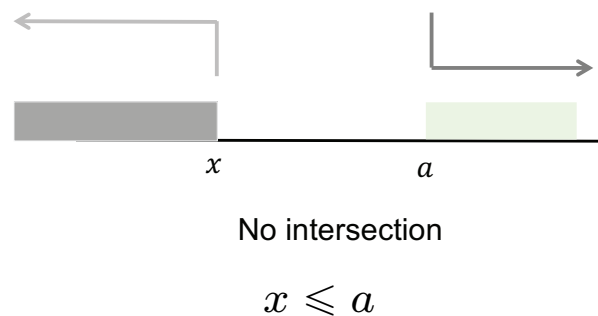
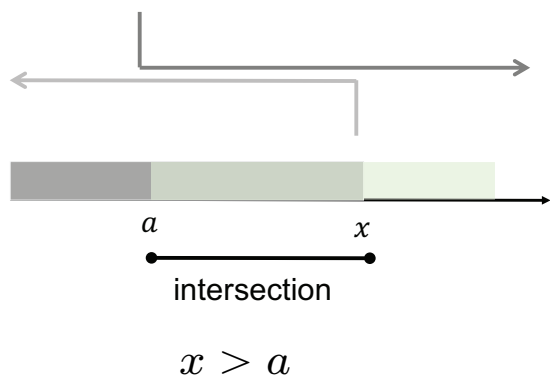
$$F_X(x | A) = F_X(x | X \leq a) = \begin{cases} 1, & x \geq a \\ \frac{1}{F_X(a)} F_X(x), & x \leq a \end{cases}$$

and conditional PDF

$$f_X(x | A) = f_X(x | X \leq a) = \frac{d}{dx} F_X(x | A) = \begin{cases} \frac{1}{F_X(a)} f_X(x), & x \leq a \\ 0, & \text{otherwise} \end{cases}$$

where $F_X(a) = \Pr(X \leq a) = \int_{-\infty}^a f_X(x) dx$ is constant for a fixed value of a

Case 2: $A = \{X > a\}$



Condition CDF

$$F_X(x | A) = F_X(x | X > a) = \frac{\Pr(X \leq x, X > a)}{\Pr(X > a)} = \begin{cases} \frac{F_X(x) - F_X(a)}{1 - F_X(a)}, & x > a \\ 0, & x \leq a \end{cases}$$

and conditional PDF

$$f_X(x | A) = f_X(x | X > a) = \frac{d}{dx} F_X(x | A) = \begin{cases} \frac{1}{1 - F_X(a)} f_X(x), & x > a \\ 0, & \text{otherwise} \end{cases}$$

Case 3: $A = \{x_1 < X \leq x_2\}$

Must consider three possible regions of x values:

(a) $x < x_1$ and $A = \{x_1 < X \leq x_2\} \Rightarrow$ no intersection

(b) $x_1 < x \leq x_2$ and $A = \{x_1 < X \leq x_2\} \Rightarrow$ intersection region is $x_1 < X \leq x$

(c) $x \geq x_2$ and $A = \{x_1 < X \leq x_2\} \Rightarrow$ intersection region is $A = \{x_1 < X \leq x_2\}$

Conditional CDF

$$\begin{aligned} F_X(x | A) &= F_X(x | x_1 < X \leq x_2) = \frac{\Pr(X \leq x, x_1 < X \leq x_2)}{\Pr(x_1 < X \leq x_2)} \\ &= \begin{cases} 0, & x < x_1 \\ \frac{F_X(x) - F_X(x_1)}{F_X(x_2) - F_X(x_1)}, & x_1 \leq x < x_2 \\ 1, & x \geq x_2 \end{cases} \end{aligned}$$

and conditional PDF

$$f_X(x | A) = f_X(x | x_1 < X \leq x_2) = \frac{d}{dx} F_X(x | A) = \begin{cases} \frac{1}{F_X(x_2) - F_X(x_1)} f_X(x), & x_1 \leq x < x_2 \\ 0, & \text{otherwise} \end{cases}$$

Example:

Consider lifetime $X \sim \exp(\lambda)$ of a machine, given that machine is working at time T . Find the PDF of X (or conditional PDF of X , given that $X > T$). This is given by

$$f_X(x \mid X \geq T) = \begin{cases} \frac{1}{1 - F_X(T)} f_X(x), & x > T \\ 0, & \text{otherwise} \end{cases}$$

Since $X \sim \exp(\lambda)$, $f_X(x) = \lambda e^{-\lambda x} u(x)$. Also,

$$1 - F_X(T) = \int_T^{\infty} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_T^{\infty} = e^{-\lambda T} \quad \Rightarrow \quad 1/(1 - F_X(T)) = e^{\lambda T}$$

So, the PDF of X , given that $X > T$, is

$$f_X(x \mid X > T) = \lambda e^{-\lambda x} e^{\lambda T}, \quad x > T$$

or

$$f_X(x \mid X > T) = \lambda e^{-\lambda(x-T)} u(x - T)$$

To compute $\Pr(A \mid X \leq x)$, use Bayes' theorem and make use of conditional CDF

Using Bayes' theorem

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}$$

Combining with conditional CDF

$$\Pr(A \mid X \leq x) = \frac{\Pr(X \leq x \mid A) \Pr(A)}{\Pr(X \leq x)} = \frac{F_X(x \mid A) \Pr(A)}{F_X(x)}$$

Example:

A resistor R has equally probable values between 50-70 Ω

Given that R is between 60-65 Ω , find the probability that $R > 62 \Omega$

If we let $A = \{R > 62\}$, then we want to find $\Pr(A \mid 60 < R \leq 65)$, where $R \sim \mathcal{U}(50, 70)$

Using Bayes' rule,

$$\Pr(A \mid 60 < R \leq 65) = \frac{\Pr(60 < R \leq 65 \mid A) \Pr(A)}{\Pr(60 < R \leq 65)} = \frac{[F_R(65 \mid A) - F_R(60 \mid A)] \Pr(A)}{F_R(65) - F_R(60)}$$

where (using Case 2)

$$F_R(r \mid A) = F_R(r \mid R > 62) = \begin{cases} \frac{F_R(r) - F_R(62)}{1 - F_R(62)} = \frac{F_R(r) - F_R(62)}{\Pr(A)}, & r > 62 \\ 0, & r \leq 62 \end{cases}$$

So $F_R(60 \mid A) = 0$ and $F_R(65 \mid A) = (F_R(65) - F_R(62))/\Pr(A)$.

Replacing this in $\Pr(A \mid 60 < R \leq 65)$, we obtain

$$\Pr(A \mid 60 < R \leq 65) = \frac{F_R(65 \mid A) \Pr(A)}{F_R(65) - F_R(60)} = \frac{\frac{(F_R(65) - F_R(62))}{\Pr(A)} \Pr(A)}{F_R(65) - F_R(60)} = \frac{F_R(65) - F_R(62)}{F_R(65) - F_R(60)}$$

where for a uniform RV $R \sim \mathcal{U}(50, 70)$, $F_R(r) = \frac{r - 50}{70 - 50} = \frac{r - 50}{20}$, for $50 < r \leq 70$

$$\Rightarrow F_R(65) = 15/20, F_R(60) = 10/20, \text{ and } F_R(62) = 12/20$$

This results in

$$\Pr(A \mid 60 < R \leq 65) = \frac{F_R(65) - F_R(62)}{F_R(65) - F_R(60)} = \frac{15/20 - 12/20}{15/20 - 10/20} = \frac{3}{5} = 0.6$$